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Integrated lithium niobate polarization beam splitter based on a photonic-crystal-assisted multimode interference coupler

MINGRUI YUAN,¹ XU HAN,¹  HUIFU XIAO,¹  THACH GIANG NGUYEN,²  ANDREAS BOES,^{2,3,4} 
GUANGHUI REN,²  QINFEN HAO,⁵ JIUZHI XUE,⁶ ARNAN MITCHELL,²  AND YONGHUI TIAN^{1,*}

¹Institute of Microelectronics and Key Laboratory for Magnetism and Magnetic Materials of MOE, School of Physical Science and Technology, Lanzhou University, Lanzhou 730000, Gansu, China

²Integrated Photonics and Applications Centre (InPAC), School of Engineering, RMIT University, Melbourne, VIC 3001, Australia

³Institute for Photonics and Advanced Sensing (IPAS), University of Adelaide, SA 5005, Australia

⁴School of Electrical and Electronic Engineering, University of Adelaide, SA 5005, Australia

⁵Institute of Computing Technology, Chinese Academy of Sciences, Beijing 100190, China

⁶Smart Liquid Crystal Technologies Company Limited, Changshu 215500, Jiangsu, China

*Corresponding author: siphoton@lzu.edu.cn

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Lithium niobate on insulator (LNOI) is a promising platform for high-speed photonic integrated circuits (PICs) that are used for communication systems due to the excellent electro-optic properties of lithium niobate (LN). In such circuits, the high-speed electro-optical modulators and switches need to be integrated with passive circuit components that are used for routing the optical signals. Polarization beam splitters (PBSs) are one of the fundamental passive circuit components for high-speed PICs that can be used to (de)multiplex two orthogonal polarization optical modes, enabling on-chip polarization division multiplexing (PDM) systems, which are suitable for enhancing the data capacity of PICs. In this Letter, we design and experimentally demonstrate a high-performance PBS constructed by a photonic crystal (PC)-assisted multimode interference (MMI) coupler. The measured polarization extinction ratio (ER) of the fabricated device is 15 dB in the wavelength range from 1525 to 1565 nm, which makes them suitable for the high-speed and large data capacity PICs required for future communication systems. © 2022 Optica Publishing Group

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Lithium niobate on insulator (LNOI) is a promising platform for realizing high-speed photonic integrated circuits (PICs) thanks to the excellent electro-optic material properties of lithium niobate (LN) [1,2]. By taking full advantage of LN, the LNOI platform enabled experimental demonstrations of high-speed and low-power electro-optic circuit components such as modulators [3,4], and high-efficiency nonlinear optical circuit components [5,6], which are enabled by the strong modal confinement and low propagation loss of LNOI [7,8]. Also, various passive circuit components have been demonstrated on the LNOI platform, such as (de)multiplexers [9,10] and filters [11,12]. Nevertheless, further expansion of the passive component toolbox is required to realize future high-performance PICs.

Polarization beam splitters (PBSs) are one of the key passive circuit components for separating and combining orthogonal polarized modes—transverse electric (TE) polarized modes and transverse magnetic (TM) polarized modes—which can be used for polarization division multiplexing (PDM) on-chip. As circuit components in PDM systems [13], PBSs provide efficient polarization management, which can be used to increase the data capacity of PICs. In the past few years, PBSs have been demonstrated with a variety of structures, including directional couplers [14,15], Mach–Zehnder interferometers [16], hybrid plasmonic waveguides [17], Bragg gratings [18], and multimode interference (MMI) couplers [19,20]. Among them, MMI couplers are attractive due to their simple fabrication and large fabrication tolerances. However, the length of a PBS based on a conventional MMI is usually a multiple of the first self-image length for both polarizations, thus resulting in a large footprint [21]. To reduce the device footprint, a two-dimensional photonic crystal (PC)—a periodic structure with exhibits a photonic bandgap—can be introduced into the MMI coupler [22]. Such PCs exhibit different transmission properties for the two polarizations of light, thanks to the inherent anisotropic structure. The different transmission properties have been used to achieve a PBS based on a PC-assisted MMI coupler—a PC-MMI PBS [23,24]. Such PBSs have two advantages: (i) the length of the MMI is not required to be an integer multiple of the self-image length for both polarizations; (ii) the performance of the device can be optimized independently for the two polarizations. Thus, it creates additional degrees of freedom to significantly reduce the size of the PBS. However, so far such a PBS has not been demonstrated on the LNOI platform.

In this Letter we propose, design, and demonstrate a high-performance PC-MMI PBS on the LNOI platform. We fabricated the proposed device by depositing a silicon nitride (Si_3N_4) layer on the top of a commercial LNOI wafer, known as Si_3N_4 -loaded LNOI or Si_3N_4 -LNOI [25,26]. Compared with PBSs based on direct etching schemes of LN [27–30], the Si_3N_4 -LNOI

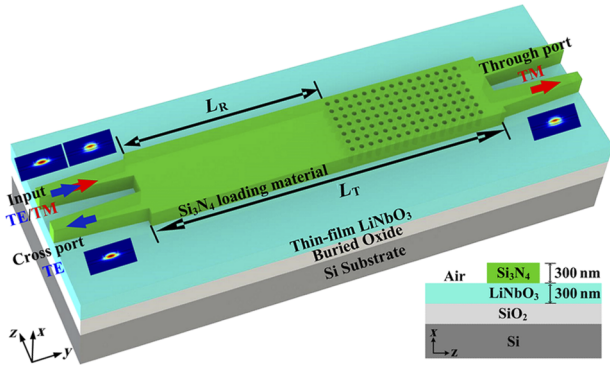


Fig. 1. Schematic diagram of the PC-MMI PBS based on the Si_3N_4 -LNOI waveguide platform. Inset shows the cross-sectional view of the waveguide.

has the advantage of near-vertical sidewalls, which is desirable for the fabrication of the PC structure and can be challenging to achieve using the LN direct etching method, which results in angled sidewalls. For both polarizations, the fabricated PC-MMI PBS achieves a polarization extinction ratio (ER) of more than 15 dB, while the insertion loss (IL) remains less than 3.86 dB in a 40-nm wavelength range from 1525 to 1565 nm. The proposed PC-MMI PBS exhibits promising performance and a compact size. This work has the potential to pave the way to realize a variety of PC-based optical components on the LNOI platform.

The 3D schematic of the proposed PC-MMI PBS, formed by a 2×2 MMI coupler with a PC structure on an Si_3N_4 -LNOI platform, is shown in Fig. 1. The working principle of the proposed device is as follows: the two-dimensional PC structure is located at the back of an MMI coupler and designed so that the PC reflects the input TE polarization light and forms the first self-image at the cross port of MMI coupler, while for the TM polarization light, the PC is transparent enabling the formation of the first self-image at the through port. As a result, the two polarizations can be effectively separated.

As a first step, we optimized the MMI coupler section of the PC-MMI PBS. To generate high-quality self-images, the width of the multimode section of the MMI coupler was set to $10 \mu\text{m}$, so that it supports more than 20 guided modes. The length of the MMI couplers is denoted as L_T . The distance between the start of the MMI coupler and the first interface of the PC is indicated as L_R . For high-quality imaging, the MMI coupler's access port is $2 \mu\text{m}$ wide to ensure that most of the input power is carried by low-order modes, and to minimize the reflection at the interfaces between the access waveguides and the MMI coupler. The center-to-center distance of access ports is set to $3.5 \mu\text{m}$ to ensure low cross talk for both polarizations. A linear cone with a $35 \mu\text{m}$ length is employed to interconnect the MMI coupler ports with the single-mode waveguide and ensure an adiabatic transition from the single-mode waveguide to the MMI coupler. The width of the single-mode waveguides is $1.20 \mu\text{m}$ for the wavelength of 1550 nm [10].

To achieve the independent optimization of both TE and TM polarizations, the correct length of L_T and L_R needs to be determined. The multimode section of the MMI coupler supports multiple guided modes, which interfere with each other according to the self-imaging effect, thus forming an N-fold image of the input field distributed along the propagation direction at periodic intervals [31]. The field distribution in the multimode

section can be written as a superposition of all the guided modes as

$$\psi(z, y) = \sum_v C_v \varphi_v(z) \exp \left[j \frac{v(v+2)\pi}{3L_\pi} y \right] \quad (1)$$

where z denotes the coordinates along the transverse axis of the MMI coupler, C_v and $\varphi_v(z)$ represent the excitation coefficients and the modal fields of the v th-order mode, respectively, and L_π represents the beat length of the two lowest-order modes of the MMI coupler, which can be expressed as

$$L_\pi = \frac{\lambda}{2(n_{\text{eff},0} - n_{\text{eff},1})} \quad (2)$$

where λ denotes the operating wavelength ($1.55 \mu\text{m}$) of the MMI coupler, and $n_{\text{eff},0}$ and $n_{\text{eff},1}$ are the effective indices of the fundamental and the first order modes of the device, respectively. Since the internal PC structure behaves as a polarization-sensitive reflector that reflects the TE polarization and transmits the TM polarization, single-fold self-images can be obtained in the forward and backward directions, respectively, by setting $L_T = L_\pi^{\text{TM}}$ and $L_R = L_\pi^{\text{TE}}/2$ [23]. By using a full vector eigenmode solver we obtained the beat lengths $L_\pi^{\text{TM}} = 157.37 \mu\text{m}$ for the TM polarization and $L_\pi^{\text{TE}} = 171.26 \mu\text{m}$ for the TE polarization. Therefore, L_T and L_R are $157.37 \mu\text{m}$ and $85.63 \mu\text{m}$, respectively.

Next, we optimized the internal PC structure. The two-dimensional PC structure consists of air holes in the Si_3N_4 thin film with a rectangular lattice, and the normal of the interface points in the Γ -X direction. In consideration of maximizing the minimum feature size, the period of the air holes is set to 420 nm and the hole radius is 100 nm to ensure an isolated feature width of 200 nm and feature spacing of 220 nm . The number of periods along the y axis is set to 150 to guarantee a reflectivity greater than 80% for the TE polarization, and the number of periods along the z axis is set to 23 to cover the entire width of the MMI coupler.

The reflection of the TE polarization and transmission of the TM polarization by the PC structure can be described in terms of a photonic energy bandgap. The simulated photon energy band for TE and TM polarizations are shown in Fig. 2, which are obtained by using a three-dimensional finite difference time domain (3D-FDTD) method and applying Bloch boundary conditions on the PC unit cell. For our device, light propagates only along the y axis (i.e., the Γ -X direction of the PC), thus only the polarization characteristics of the photonic bandgap in the

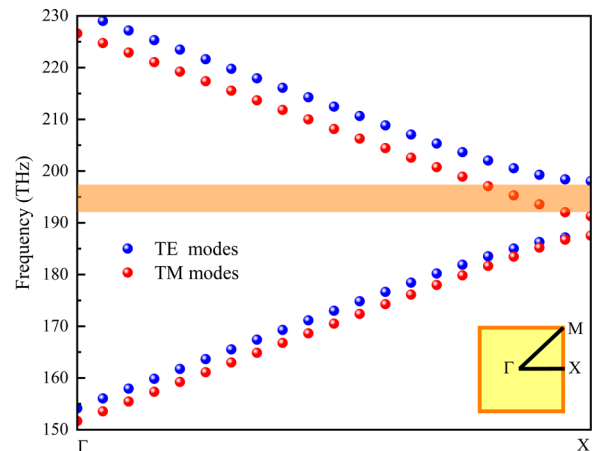


Fig. 2. Simulated band structure of the proposed two-dimensional PC structure for TE and TM polarizations.

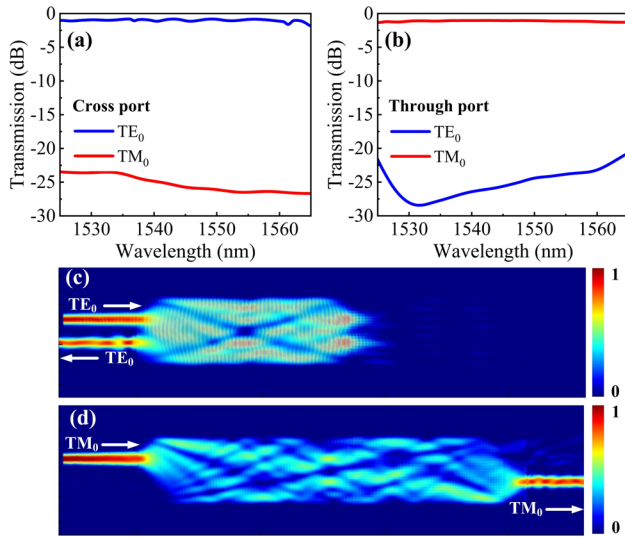


Fig. 3. Simulated normalized transmission spectra for TE₀ and TM₀ polarizations at (a) cross port and (b) through port at the wavelength range of 1525–1565 nm. Simulated E-field profiles at 1550 nm for (c) TE₀ and (d) TM₀ input into the PC-MMI PBS

Γ -X direction need to be considered. From Fig. 2, it can be seen that the photon bandgap of the TE polarization has a frequency range of 187.6–198.0 THz (corresponding to an 84-nm wavelength range from 1515.1 to 1599.1 nm), whereas the TM polarization has a narrower photon bandgap with a frequency in the range of 187.6–191.3 THz (corresponding to a 31-nm wavelength range from 1568.2 to 1599.1 nm). To separate the two polarizations, the PC structure should have the transmission and reflection coefficients of $t_{TE} \approx 0$, $r_{TE} \approx 1$ and $t_{TM} \approx 1$, $r_{TM} \approx 0$ for TE and TM polarizations, which is satisfied in the wavelength range of 1515.1 to 1568.2 nm (indicated by the orange area in Fig. 2).

So far, we have designed the PC-MMI PBS by designing the MMI coupler and PC structure separately. Next, we performed the 3D-FDTD method on the combined device to optimize the structure and evaluate its performance. Figures 3(a) and 3(b) show the simulated normalized transmission spectra for the two polarizations (at the cross port and the through port) as a function of wavelength from 1525 to 1565 nm. It can be seen that the ILs of the TE polarization and the TM polarization within a wavelength range of 40 nm are less than 1.86 dB and 1.34 dB, respectively. The ERs of the TE polarization and the TM polarization in the wavelength range from 1525 to 1565 nm are better than 22.5 dB and 19.5 dB, respectively. Figures 3(c) and 3(d) show the corresponding E-field profiles at a wavelength of 1550 nm. From the figures, one can see that the TE polarization is reflected to the cross port, while the TM polarization is transmitted to the through port. This shows that the designed PC-MMI PBS device can be used to separate the two polarizations effectively.

Next, we fabricated the proposed PC-MMI PBS on an X-cut LNOI wafer provided by “NANOLN”. By using reactive sputtering, a 300-nm-thick Si₃N₄ thin film is deposited on the LNOI wafer. After that, electron beam lithography (EBL) and inductively coupled plasma (ICP) etching processes were used to pattern the Si₃N₄-LNOI waveguides as well as the grating couplers to interface the device with standard single-mode fibers. Three polarization splitter-rotators (PSRs) based on a cascade of

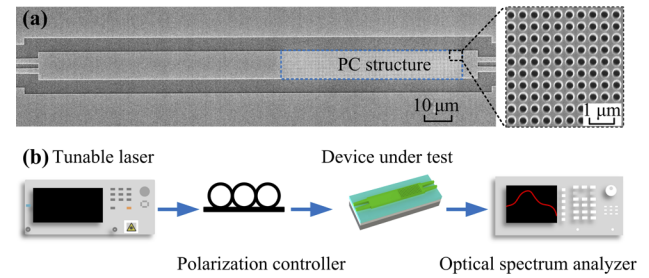


Fig. 4. (a) SEM image of the fabricated PC-MMI PBS. The inset shows the detail of the PC structure. (b) Experimental setup for characterizing the proposed device.

asymmetric directional couplers (ADCs) were integrated after the PC-MMI PBS to characterize the performance of the different polarization channels. The detailed design and performance of the ADC PSR can be found in our previous work [10]. In addition, two connected reference PSRs were fabricated next to the device to eliminate their impact on the performance of the device. The footprint of the fabricated MMI is $10 \times 157.4 \mu\text{m}^2$, with a minimum feature size of 200 nm for the PC structure. Figure 4(a) shows a scanning electron microscopy (SEM) image of the fabricated PC-MMI PBS, and the inset shows the detail of the PC structure.

To characterize the performance of the fabricated PC-MMI PBS, we used an experimental setup that included a tunable laser (TL), a polarization controller, and an optical spectrum analyzer (OSA), as shown in the schematic diagram of Fig. 4(b). In the characterization process, the laser output from the TL was connected to the polarization controller, subsequently coupled into and out of the chip via grating couplers, and then connected to an OSA to record the transmitted spectrum. The measured transmission spectra were normalized by the measured responses of the grating couplers and the PSRs to reflect the response of the investigated PC-MMI PBS themselves.

Figures 5(a) and 5(b) show the normalized spectra of the fabricated PC-MMI PBS measured at the cross port and the through port, respectively. In the wavelength range from 1525 to 1565 nm, the fabricated PBS achieved a measured ER of more than 15 dB for both polarizations, while the IL remains below 3.86 dB. At a wavelength of 1550 nm, the ILs are 2.16 dB and 2.53 dB, while the ERs are 25.05 dB and 25.26 dB, for the TE and TM polarizations, respectively. The results show that the device exhibits promising performance for future PDM techniques on the LNOI platform.

We provide a brief summary of previously reported PBSs on the LNOI platform in Table 1. It can be seen that the presented work is the first experimental demonstration of a PBS on the LNOI platform with low IL and cross talk.

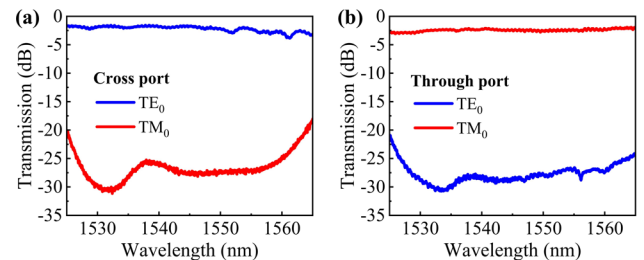


Fig. 5. Measured transmission spectra of TE₀ and TM₀ polarizations at (a) cross port and (b) through port.

Table 1. Comparison of Some PBSs Reported on LNOI Platforms

Ref.	Function	Length (μm)	ER (dB)	IL (dB)	Bandwidth (nm)
27	Simulated	~430	47.7 (TE)	0.6 (TE)	> 200 (ER > 17.8 dB, TE)
			48.0 (TM)	0.9 (TM)	> 200 (ER > 22.8 dB, TM)
28	Simulated	160	~24 (TE)	0.3 (TE)	185 (ER > 20 dB, TE)
			~25 (TM)	1.0 (TM)	85 (ER > 20 dB, TM)
29	Simulated	16	26.7 (TE)	< 0.05 (TE&TM)	> 150 (IL < 0.5 dB, TE)
			21.3 (TM)		> 200 (IL < 0.2 dB, TM)
30	Simulated	61.5	22.82 (TE)	0.12 (TE)	38 (ER > 20 dB & CT < -20 dB)
			26.15 (TM)	0.47 (TM)	40
This work	Simulated	157.4	25.3 (TE)	0.89 (TE)	(ER > 22.5 dB & IL < 1.86 dB, TE)
			23.4 (TM)	1.06 (TE)	40 (ER > 19.5 dB & IL < 1.34 dB, TM)
			25.05 (TE)	2.16 (TE)	40 (ER > 15 dB & IL < 3.86 dB)
	Experimental		25.26 (TM)	2.53 (TM)	

In summary, we proposed and experimentally demonstrated a high-performance PBS based on a PC-assisted MMI coupler on the LNOI platform. Thanks to the different transmission characteristics of the PC structure for the two polarizations of light, the length of the MMI is not required to be an integer multiple of the self-imaging length for both polarizations, thus reducing the size of the MMI significantly ($10 \times 157.4 \mu\text{m}^2$). Moreover, the performance of the device can be optimized independently for the two polarizations. The ILs of the fabricated PC-MMI PBS are 2.16 dB and 2.53 dB, while the ERs are 25.05 dB and 25.26 dB, for the TE polarization and the TM polarization, respectively, at a wavelength of 1550 nm. For both polarizations, the device achieves ERs above 15 dB and ILs below 3.86 dB in the 1525 to 1565 nm wavelength range. The superior performance and compact size make the device a very attractive candidate for high-speed and high-capacity LNOI PDM systems. This also paves the way for the implementation of various PC-based optical components on the LNOI platform.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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